WORKED SOLUTION

STEP III 1991, Question 14

Mechanics • Circular Motion with Springs

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Read the question before you go on. It is in the STEP Support Programme's question database:

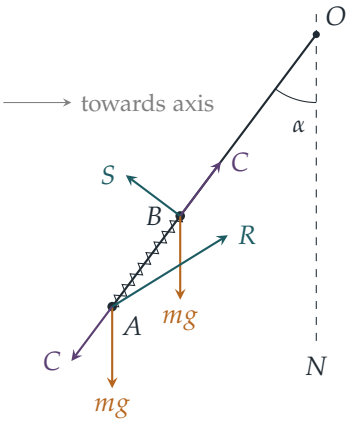
step.maths.org/questions/91-s3-q14

The key idea

Everything moves in a horizontal circle about ON at the same angular speed, so each mass has a known acceleration $\omega^2 r$ towards the axis and nothing else. For the ring, which sits on a *smooth* rod, the cleanest equation is the one along the rod, because the unknown normal reaction drops out of it. Then the spring force links the ring to the particle at A, where the direction of the rod's reaction pins down k.

Solution

Model. The rod is light and smooth, A and B are particles of mass m, and the spring lies along the rod between B and A. Write $\alpha = \sin^{-1}\frac{3}{5}$, so $\sin\alpha = \frac{3}{5}$, $\cos\alpha = \frac{4}{5}$, and $\omega^2 = \frac{5g}{2a}$. Let $OB = x$; then $AB = 2a - x$.



Here C is the thrust in the spring (we shall find that the spring is *compressed*, so it pushes B towards O and pushes A away from O), S is the normal reaction of the smooth rod on B (perpendicular to the rod, in the vertical plane, sense to be found), and R is

the reaction of the rod on A. Since ω is constant and B is at rest relative to the rod, there is no acceleration across the vertical plane, so all the forces lie in that plane.

The length OB

B moves in a horizontal circle of radius $x\sin\alpha$, so its acceleration is $\omega^2 x\sin\alpha$ horizontally towards ON. Resolve along the rod in the direction $O \to A$ (down and away from the axis). The acceleration component in this direction is $-\omega^2 x\sin\alpha\cdot\sin\alpha$, since the acceleration points towards the axis and the rod makes angle α with the vertical.

It is simplest to let the spring force on B be a *tension* T (pulling B towards A) and let the sign sort itself out. By Hooke's law,

$$T = \frac{kmg\left((2a-x)-a\right)}{a} = \frac{kmg(a-x)}{a}.$$

The reaction S has no component along the rod because the rod is smooth. So, along OA,

$$mg\cos\alpha + \frac{kmg(a-x)}{a} = -m\omega^2 x\sin^2\alpha.$$

With $\cos\alpha = \frac{4}{5}$, $\sin^2\alpha = \frac{9}{25}$ and $\omega^2 = \frac{5g}{2a}$, the right-hand side is $-\frac{9gx}{10a}\,m$. Dividing by mg and multiplying by $10a$:

$$\begin{aligned} 8a + 10k(a-x) &= -9x \\ (10k+8)a &= (10k-9)x, \end{aligned}$$

so

$$OB = x = \frac{(10k+8)a}{10k-9},$$

as required. Notice that $x > a$ (we need $10k > 9$ for a positive answer), so $AB = 2a - x < a$: the spring is compressed, with thrust

$$C = -T = \frac{kmg(x-a)}{a} = \frac{17k\,mg}{10k-9},$$

using $x - a = \dfrac{17a}{10k-9}$.

Finding k

Ask yourself Why isn't the reaction at A perpendicular to the rod, like the one at B?

The particle at A is *attached* to the rod, not threaded on it, so the rod can push or pull it in any direction. Smoothness only restricts the force on the sliding ring. So we write R in components: R_h horizontally towards the axis and R_v vertically upwards.

A moves in a circle of radius $2a\sin\alpha = \frac{6a}{5}$, so its acceleration is $\omega^2\cdot\frac{6a}{5} = 3g$ towards the axis. The thrust C acts on A along OA away from O, so it has components $C\sin\alpha$

away from the axis and $C\cos\alpha$ downwards. Hence

$$\text{horizontal (towards axis):}\quad R_h - \tfrac{3}{5}C = 3mg,$$
$$\text{vertical:}\quad R_v - mg - \tfrac{4}{5}C = 0.$$

Put $C = c\,mg$. The reaction makes angle $\tan^{-1}\frac{13}{21}$ with the horizontal, so

$$\frac{R_v}{R_h} = \frac{1+\frac{4}{5}c}{3+\frac{3}{5}c} = \frac{13}{21} \implies 21 + \tfrac{84}{5}c = 39 + \tfrac{39}{5}c \implies 9c = 18, \quad c = 2.$$

(Both R_h and R_v are positive for $c > 0$, so the reaction really does point upwards and inwards and there is no sign ambiguity.) Then

$$\frac{17k}{10k-9} = 2 \implies 17k = 20k - 18 \implies k = 6.$$

So $k = 6$, giving $OB = \frac{68a}{51} = \frac{4}{3}a$, which is safely between a and $2a$, and a thrust $C = 2mg$. (The reaction on A is then $R_h = \frac{21}{5}mg$, $R_v = \frac{13}{5}mg$.)

The reaction on the ring

Ask yourself Which direction should we resolve in to get S with the least work?

Perpendicular to the rod: the spring force then drops out. Take the unit direction perpendicular to OA in the vertical plane pointing *upwards and away from the axis*; it makes angle α with the horizontal, so it has components $\cos\alpha$ outwards and $\sin\alpha$ upwards. Let S be the component of the reaction on B in this direction. The weight contributes $-mg\sin\alpha$, and the acceleration ($\omega^2 x\sin\alpha$ towards the axis) contributes $-\omega^2 x\sin\alpha\cos\alpha$. So

$$S - mg\sin\alpha = -m\omega^2 x\sin\alpha\cos\alpha.$$

With $x = \frac{4}{3}a$:

$$m\omega^2 x\sin\alpha\cos\alpha = m\cdot\frac{5g}{2a}\cdot\frac{4a}{3}\cdot\frac{12}{25} = \frac{8}{5}mg, \qquad \text{so} \qquad S = \frac{3}{5}mg - \frac{8}{5}mg = -mg.$$

The reaction between the rod and the ring therefore has magnitude

$$\boxed{mg},$$

and the minus sign tells us it acts perpendicular to the rod, *downwards and towards the axis*.

Checks

- Full vector check on B (horizontal components towards the axis, vertical upwards), with $C = 2mg$ acting along AO and $S = mg$ as found. Horizontal: $\frac{3}{5}(2mg) + \frac{4}{5}(mg) = 2mg$, and $m\omega^2 x\sin\alpha = m\cdot\frac{5g}{2a}\cdot\frac{4a}{3}\cdot\frac{3}{5} = 2mg$. Vertical: $\frac{4}{5}(2mg) - \frac{3}{5}mg - mg = 0$. Both balance.

- As $k \to \infty$ the spring becomes rigid and $OB \to a$, its natural length, as it should.
- Numerically, with $m = g = a = 1$, we checked the OB formula exactly for 200 random values of k, found the angle $\tan^{-1}\frac{13}{21}$ only at $k = 6$, and got $|S| = 1.0$. Stepping B's motion along the rod from a small displacement keeps it near $x \approx 1.3333$.

Common slips

- Assuming the spring is stretched and then being puzzled by $x > a$. Let the sign of the tension come out of the algebra.
- Taking the reaction at A to be perpendicular to the rod. Only the sliding ring has that restriction.
- Using radius x instead of $x\sin\alpha$ for the circular motion.

Beyond STEP

In a frame rotating with the rod, B feels a "centrifugal" force $m\omega^2 r$ outwards, and the along-rod equation becomes an ordinary equilibrium problem. Taking that viewpoint further gives effective potentials, the Coriolis force (which here is balanced by the rod's reaction as soon as B moves along the rod), and the stability of the equilibrium: is $x = \frac{4}{3}a$ stable? Kleppner and Kolenkow, *An Introduction to Mechanics*, treats non-inertial frames and rotating systems at exactly this level.

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